

Complex Analysis Homework 4

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2. Let u be a harmonic function in the unit disc that is continuous on its closure. Deduce Poisson's integral formula

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - |z_0|^2}{|e^{i\theta} - z_0|^2} u(e^{i\theta}) d\theta \quad \text{for } |z_0| < 1$$

from the special case $z_0 = 0$ (the mean value theorem). Show that if $z_0 = re^{i\varphi}$, then

$$\frac{1 - |z_0|^2}{|e^{i\theta} - z_0|^2} = \frac{1 - r^2}{1 - 2r \cos(\theta - \varphi) + r^2} = P_r(\theta - \varphi),$$

and we recover the expression for the Poisson kernel derived in the exercises of the previous chapter.

[Hint: Set $u_0(z) = u(T(z))$ where

$$T(z) = \frac{z_0 - z}{1 - \overline{z_0}z}.$$

Prove that u_0 is harmonic. Then apply the mean value theorem to u_0 , and make a change of variables in the integral.]

证明 根据第八章引理 1.3 可知 u_0 也是调和的

$$u_0(0) = u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u_0(e^{i\varphi}) d\varphi.$$

作换元 $T(e^{i\varphi}) = e^{i\theta}$, 即有

$$e^{i\varphi} = T(e^{i\theta}) = \frac{z_0 - e^{i\theta}}{1 - \overline{z_0}e^{i\theta}},$$

两边求导

$$\begin{aligned} d\varphi &= \frac{(|z_0|^2 - 1)e^{i\theta}}{(1 - \bar{z}_0 e^{i\theta})^2} e^{-i\varphi} d\theta = \frac{(|z_0|^2 - 1)e^{i\theta}}{(1 - \bar{z}_0 e^{i\theta})(z_0 - e^{i\theta})} d\theta \\ &= \frac{(|z_0|^2 - 1)}{(e^{-i\theta} - \bar{z}_0)(z_0 - e^{i\theta})} d\theta = \frac{1 - |z_0|^2}{|e^{i\theta} - z_0|^2} d\theta, \end{aligned}$$

代入即得

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - |z_0|^2}{|e^{i\theta} - z_0|^2} u(e^{i\theta}) d\theta.$$

1. A holomorphic mapping $f : U \rightarrow V$ is a local bijection on U if for every $z \in U$ there exists an open disc $D \subset U$ centered at z , so that $f : D \rightarrow f(D)$ is a bijection.

Prove that a holomorphic map $f : U \rightarrow V$ is a local bijection on U if and only if $f'(z) \neq 0$ for all $z \in U$.

[Hint: Use Rouché's theorem as in the proof of Proposition 1.1.]

证明 必要性：根据命题 1.1 即得

充分性：若 $f'(z_0) \neq 0$ ，不妨设 $f(z_0) = w_0$ ，从而存在 $B(z_0, \delta)$ 使得 $f(z) \neq f(z_0), z \in B(z_0, \delta)$ 且 $z \neq z_0$ ，任取 $|w| < w_m = \inf_{|z-z_0|=\delta} |f(z)-w_0|$ ，于是考虑 $g(z) = f(z) - w$ ，根据 Rouché's theorem 有 $g(z)$ 在 $B(z_0, \delta)$ 上只有一个零点，从而 f 是从 $B(z_0, \delta)$ 到 $f(B(z_0, \delta)) = B(w_0, w_m)$ 的双射

2. Suppose $F(z)$ is holomorphic near $z = z_0$ and $F(z_0) = F'(z_0) = 0$, while $F''(z_0) \neq 0$. Show that there are two curves Γ_1 and Γ_2 that pass through z_0 , are orthogonal at z_0 , and so that F restricted to Γ_1 is real and has a minimum at z_0 , while F restricted to Γ_2 is also real but has a maximum at z_0 .

[Hint: Write $F(z) = (g(z))^2$ for z near z_0 , and consider the mapping $z \mapsto g(z)$ and its inverse.]

证明 设 g 使得在 z_0 的邻域上有 $F(z) = (g(z))^2$ ，从而可设

$$g(z) = a(z - z_0) + \cdots$$

根据前一题结论，存在 δ 使得 g 在 $B(0, \delta)$ 上是单射且与 $B(0, w_m)$ 是共形

考虑 g^{-1} ，在实轴和虚轴下的像，分别设为 Γ_1, Γ_2 ，于是在 Γ_1 上 $f \geq 0$ ，在 Γ_2 上 $f \leq 0$ ，且极值点恰好在 $z = z_0$ 处取到。并且根据共形的性质，这两条曲线在 z_0 处仍然正交

3. Suppose U and V are conformally equivalent. Prove that if U is simply connected, then so is V . Note that this conclusion remains valid if we merely assume that there exists a continuous bijection between U and V .

证明 由于 U 和 V 是共形的, 所以存在全纯的双射 $f: U \rightarrow V$

任取 V 中一条闭曲线 $\gamma: [0, 1] \rightarrow V$, 令

$$\alpha = f^{-1} \circ \gamma: [0, 1] \rightarrow U.$$

则 α 是 U 中的一条闭曲线。

因为 U 单连通, α 可在 U 中连续收缩到一点, 即存在连续映射

$$H: [0, 1] \times [0, 1] \rightarrow U$$

满足

$$H(t, 0) = \alpha(t), \quad H(t, 1) = z_0,$$

$$H(0, s) = H(1, s) = z_0,$$

其中 z_0 为常数点。

令 $\tilde{H}(t, s) = f(H(t, s))$, 则 $\tilde{H}$ 连续, 且

$$\tilde{H}(t, 0) = \gamma(t), \quad \tilde{H}(t, 1) = f(z_0),$$

$$\tilde{H}(0, s) = \tilde{H}(1, s) = f(z_0).$$

因此 γ 可在 V 中收缩为一点。由 γ 的任意性, V 单连通。

4. Does there exist a holomorphic surjection from the unit disc to $\mathbb{C}$?

[Hint: Move the upper half-plane “down” and then square it to get $\mathbb{C}$.]

证明 考虑 $f = (i\frac{1-z}{1+z} - i)^2$, 即将单位圆盘映射到上半平面, 向下平移后再平方。

5. Prove that $f(z) = -\frac{1}{2}(z + \frac{1}{z})$ is a conformal map from the half-disc $\{z = x + iy : |z| < 1, y > 0\}$ to the upper half-plane.

[Hint: The equation $f(z) = w$ reduces to the quadratic equation $z^2 + 2wz + 1 = 0$, which has two distinct roots in $\mathbb{C}$ whenever $w \neq \pm 1$. This is certainly the case if $w \in \mathbb{H}$.]

证明 设 $z = x + iy$, 于是

$$x + iy + \frac{1}{x + iy} = x + \frac{x}{x^2 + y^2} + i(y - \frac{y}{x^2 + y^2})$$

$y - \frac{y}{x^2 + y^2} < 0$, 于是像在上半平面中

对于上半平面中的任何一点 w , 不妨设 $f(z) = w$, 从而

$$z^2 + 2zw + 1 = 0$$

于是对于两个根有

$$z_1 z_2 = 1, z_1 + z_2 = -2w$$

不妨设 $|z_1| < 1 < |z_2|$, 于是 $\operatorname{Im} z_1 + \operatorname{Im} z_2 = \operatorname{Im} z_1(1 - \frac{1}{|z_1|^2}) < 0$, 从而

$\operatorname{Im} z_1 > 0$, 即 z_1 属于上半圆盘

而 $f'(z) = -\frac{1}{2}(1 - \frac{1}{z^2}) \neq 0$, 综上所述有 f is a conformal map from the half-disc $\{z = x + iy : |z| < 1, y > 0\}$ to the upper half-plane.

9. Prove that the function u defined by

$$u(x, y) = \operatorname{Re} \left(\frac{i + z}{i - z} \right) \quad \text{and} \quad u(0, 1) = 0$$

is harmonic in the unit disc and vanishes on its boundary. Note that u is not bounded in $\mathbb{D}$.

证明 令 $f(z) = \frac{i + z}{i - z}$. 在单位圆盘 $\mathbb{D} = \{z : |z| < 1\}$ 内, 分母 $i - z \neq 0$, $f(z)$ 是 $\mathbb{D}$ 内的全纯函数. 因此 $u = \Re(f(z))$ 在 $\mathbb{D}$ 内调和.

设 $|z| = 1$, 则 $z\bar{z} = 1$, 即 $\bar{z} = \frac{1}{z}$.

$$\frac{i + z}{i - z} = \frac{(i + z)\overline{(i - z)}}{|i - z|^2} = \frac{(i + z)(-i - \bar{z})}{|i - z|^2}$$

计算分子:

$$(i + z)(-i - \bar{z}) = 1 - i\bar{z} + iz - z\bar{z} = i(z - \bar{z})$$

该结果实部为 0, 即:

$$u(x, y) = \Re \left(\frac{i + z}{i - z} \right) = 0, \quad \forall |z| = 1.$$

取沿虚轴逼近边界的点列: $z = iy$, 其中 $0 < y < 1$.

$$u(0, y) = \Re \left(\frac{i + iy}{i - iy} \right) = \Re \left(\frac{i(1 + y)}{i(1 - y)} \right) = \Re \left(\frac{1 + y}{1 - y} \right) = \frac{1 + y}{1 - y}$$

当 $y \rightarrow 1^-$ 时,

$$u(0, y) = \frac{1+y}{1-y} \rightarrow +\infty$$

故函数 $u(x, y)$ 在单位圆盘 $\mathbb{D}$ 内无界。